

## **Meta-analysis Appraisal Tool for Environmental Sciences (MATES) task feedback**

### ***MATES checklist***

The MATES checklist provides a point-by-point appraisal of the reporting of methodological items in meta-analyses. To apply MATES to a meta-analysis, check whether each of the following items is present. Some MATES items have 'Not Applicable' criteria if a previous question needs to be reported or, a particular caveat makes an item not appropriate for assessment. For any studies that include multiple separate meta-analyses, complete the reporting appraisal for the first analysis (i.e., the analysis that comes first in the study being appraised). Please read the glossary of terms at the end of the checklist.

### ***MATES item ~~15~~: Effect size statistic formula reporting***

#### **Question**

- *Do the authors report the formula (or reference for formula) for the effect size statistic/measure used (e.g.,  $Z_r$ , Hedge's  $g$ , Cohen's  $d$ , or  $\ln RR$ ) in the manuscript or supplementary material?*

#### **Criteria for 'Yes'**

- *The authors have reported the formula (or reference for the formula) for the effect size statistic/measure used in the manuscript or supplementary material.*

#### **Criteria for 'No'**

- *The authors have not reported the formula (or reference for the formula) for the effect size statistic/measure used in the manuscript or supplementary material.*

#### Criteria for 'Not Applicable'

- ~~There is no 'Not Applicable' option for MATES item 5.~~ The authors conducted an arms-based meta-analysis or extracted the effect sizes directly from the included primary studies.

#### **MATES item 26: Model type reporting**

**Commented [KM1]:** A few suggested the specifications is more important - we had discussed this before.

#### Question

- Do the authors report the type of the meta-analytic/meta-regression model (e.g., fixed, random, or multi-level models), either descriptively or in statistical notation in the manuscript or supplementary material?

#### Criteria for 'Yes'

- The authors have reported the type of the meta-analytic/meta-regression model, either descriptively or in statistical notation in the manuscript or supplementary material.

#### Criteria for 'No'

- The authors have **not** reported the type of the meta-analytic/meta-regression model, either descriptively or in statistical notation in the manuscript or supplementary material.

#### Criteria for 'Not Applicable'

There is no 'Not Applicable' option for MATES item 26.

**MATES item 38: Effect size dependence (non-independence) reporting**

Question

- Do the authors report the dependence (e.g., multiplicity or, repeated measure) of effect size estimates (e.g., acknowledge multiple effect sizes were taken from a single study) in the manuscript or supplementary material?

Criteria for 'Yes'

- The authors have reported whether there is dependence of effect size estimates in the manuscript or supplementary material. ~~If a meta-analysis reports that effect size estimates are assumed to be independent allocate 'Yes'.~~ Alternatively, the authors may report effect size estimates are assumed to be independent.

**Commented [KM2]:** Should this actually be non-applicable

Criteria for 'No'

- The authors have **not** reported whether there is dependence of effect size estimates in the manuscript or supplementary material.

Criteria for 'Not Applicable'

- ~~The authors extracted one effect size per each primary study. There is no 'Not Applicable' option for MATES item 8.~~

**MATES item 41: Software reporting**

Question

- Do the authors report the name, or a reference to the software or programming language used to conduct the analysis (i.e., used to conduct the computations,

*statistical models, and inferences comprising the meta-analysis) in the manuscript or supplementary material?*

**Criteria for 'Yes'**

- *The authors have reported the name, or a reference to the software or programming language used to conduct the analysis in the manuscript or supplementary material.*

**Criteria for 'No'**

- *The authors have reported the name, or a reference to the software or programming language used to conduct the analysis in the manuscript or supplementary material.*
- **Criteria for 'Not Applicable'**

*There is no 'Not Applicable' option for MATES item 4.*

**MATES item ~~5~~3: Number of included primary studies reporting**

**Question**

- *Do the authors report the total number of primary studies (i.e., separate individual articles) included in the meta-analysis in the manuscript or supplementary material?*

**Criteria for 'Yes'**

- *The authors have reported the total number of primary studies synthesised in the meta-analysis in the manuscript or supplementary material. The total number of studies can be stated ~~inferred from~~ a screening flowchart (e.g., PRISMA flowchart) or the authors can state the number of studies per subgroup analysis.*

#### Criteria for 'No'

- The authors have **not** reported the total number of primary studies synthesised in the meta-analysis in the manuscript or supplementary material. The total number of studies cannot be inferred by counting from a forest plot, or from a provided dataset.

#### Criteria for 'Not Applicable'

- There is no 'Not Applicable' option for MATES item ~~53~~.

#### **MATES item ~~64~~: Number of included effect sizes reporting**

##### Question

- Do the authors report the total number of effect size estimates synthesised in the meta-analysis in the manuscript or supplementary material?

#### Criteria for 'Yes'

- The authors have reported the total number of effect sizes estimates synthesised in the meta-analysis in the manuscript or supplementary material. ~~The total number of effect sizes cannot be inferred by counting and must be stated explicitly.~~

#### Criteria for 'No'

- The authors have not reported the total number of effect sizes synthesised in the meta-analysis in the manuscript or supplementary material. ~~The total number of effect sizes cannot be inferred by counting and must be stated explicitly.~~ Alternatively, the total number of effect sizes can only be inferred by counting.

#### Criteria for 'Not Applicable'

- There is no 'Not Applicable' option for MATES item 64.

#### **MATES item 7: Overall-Pooled mean effect reporting**

##### Question

- Do the authors report the overall-pooled mean effect of the meta-analysis/meta-regression using a plot, table or within text in the manuscript or supplementary material?

##### Criteria for 'Yes'

- The authors reported the overall-pooled mean effect of the meta-analysis/meta-regression in the text, or within a plot or table in the manuscript or supplementary material. The study may have reported a pooled mean effect for each subgroup analysis.

##### Criteria for 'No'

- The authors have not reported the overall-pooled mean effect of the meta-analysis/meta-regression in the text, or within a plot or table in the manuscript or supplementary material.

##### Criteria for 'Not Applicable'

- ⊖ There is no 'Not Applicable' option for MATES item 8.

**MATES item 87: Effect size weightings strategy reporting**

**Commented [KM3]:** Difficult to address the comments discuss with Shinichi and others.

Question

- Do the authors report ~~the strategy used to assign weights~~~~the sampling variance, sampling error or equivalent~~ (e.g., the sampling variance, sampling error or equivalent i.e., strategy to assign weights) of effect size estimates in the manuscript or supplementary material?

Criteria for 'Yes'

- The authors have reported ~~the strategy used to assign weights~~ ~~sampling variance, sampling error or equivalent~~ of effect size estimates in the manuscript or supplementary material. The authors must report the weighting strategy and not solely the weights of each study.

Criteria for 'No'

- The authors have **not** reported the ~~strategy used to assign weights~~ ~~sampling variance, sampling error or equivalent~~ of effect size estimates in the manuscript or supplementary material.

Criteria for 'Not Applicable'

- The authors have conducted a meta-analysis without weights (i.e., unweighted meta-analysis).

**MATES item 910: Heterogeneity reporting**

Question

- Do the authors report an assessment of heterogeneity (e.g.,  $I^2$ ,  $\tau^2$ ,  $\sigma^2$  or, Q-statistic) between effect size estimates for the primary or first analysis in the manuscript or supplementary material?

Criteria for 'Yes'

- The authors have reported an assessment of heterogeneity between effect size estimates for the primary or first analysis in the manuscript or supplementary material.

Criteria for 'No'

- The authors have **not** reported an assessment of heterogeneity between effect size estimates for the primary or first analysis in the manuscript or supplementary material.

Criteria for 'Not Applicable'

- The study has stated that the analysis is unweighted.

**MATES item 101: Publication bias reporting**

Question

- Do the authors report a ~~an~~ statistical or graphical assessment of publication bias in the manuscript or supplementary material?

Criteria for 'Yes'



- The authors have reported a statistical or graphical assessment of publication bias in the manuscript or supplementary material.

#### Criteria for 'No'

- The authors have **not** reported a statistical or graphical assessment of publication bias in the manuscript or supplementary material.

#### Criteria for 'Not Applicable'

- The authors report that an assessment of publication bias cannot be conducted.

### **MATES item 112: Sensitivity analysis reporting**

#### Question

- Do the authors report a sensitivity analysis or subset analysis in addition to the publication bias assessment (e.g., leave-one-out analysis, use of alternative model or within-study risk of bias assessment) in the manuscript or supplementary material?

#### Criteria for 'Yes'

- The authors reported a sensitivity analysis in addition to the publication bias assessment in the manuscript or supplementary material.

#### Criteria for 'No'

- The authors have not reported a sensitivity analysis in addition to the publication bias assessment in the manuscript or supplementary material.

#### Criteria for 'Not Applicable'

- There is no 'Not Applicable' option for MATES item 12.

### **MATES item 12: Code reporting**

#### Question

- Do the authors report any relevant code used to conduct the analysis (i.e., used to conduct the computations, statistical models, and inferences comprising the meta-analysis) in the manuscript, supplementary material, or external repository?

#### Criteria for 'Yes'

- The authors have reported any relevant code used to conduct the analysis in the manuscript, supplementary material, or external repository. Common examples of software used to perform meta-analysis, which can generate code or syntax, include R, Python, SPSS, and STATA.

#### Criteria for 'No'

- The authors have **not** reported any relevant code used to conduct the analysis in the manuscript, supplementary material, or external repository. Common examples of software used to perform meta-analysis, which can generate code or syntax, include R, Python, SPSS, and STATA.

#### Criteria for 'Not Applicable'

- The authors have used a GUI/click based software that does not produce a code or syntax (e.g., comprehensive meta-analysis software). Alternatively, the authors have **not** reported the software used to conduct the analysis (i.e., 'No' to MATES item 44).

**Commented [KM4]:** A flaw of the reordering makes some questions such as this not beside the referenced item.

### **MATES item 13: Data reporting/archiving**

#### Question

- Do the authors report the raw data (i.e., used to calculate effect sizes and conduct the meta-analysis) extracted from each eligible primary study (i.e., research paper) in the manuscript, supplementary material, or external repository?

#### Criteria for 'Yes'

- The authors have reported the raw data extracted from each eligible primary in the manuscript, supplementary material, or external repository. Alternatively, the authors may state that the raw data is sensitive and cannot be shared. In cases when the authors state the data is sensitive and cannot be shared allocate 'Yes'.

Commented [KM5]: Should this be allocated NA?

#### Criteria for 'No'

- The authors have not reported raw the data extracted from each eligible primary study the manuscript, supplementary material, or external repository. Alternatively, the authors may state the raw data is available upon request. In cases when the authors state data is available upon request allocate 'No'.

#### Criteria for 'Not Applicable'

- There is no 'Not Applicable' option for MATES item 13.

### **MATES item 14: Variable description reporting**

#### Question

- Do the authors report descriptions of the variables extracted (i.e., meta-data or columns in dataset) from each eligible study primary study (i.e., research paper) in the manuscript, supplementary material, or external repository?

Criteria for 'Yes'

- The authors have reported descriptions of the variables extracted from each eligible study primary study in the manuscript, supplementary material, or external repository.

Criteria for 'No'

- The authors have **not** reported descriptions of the variables extracted from each eligible study primary study in the manuscript, supplementary material, or external repository.

Criteria for 'Not Applicable'

- ~~The authors have **not** reported the data extracted from each eligible primary study in the manuscript, supplementary material, or external repository (i.e., 'No' for MATES item 13) or I~~the authors have reported that the data cannot be provided because the data is sensitive.